

Logical actions. Let $(G, \circ)$ be a finite group with identity e_G . The *logical value* (or simply *logic*) of a reasoning chain is an element of an abstract space $\mathcal{R}$ equipped with a G -action $r \mapsto g \cdot r$. Each edge $e \in E$ is assigned a *logical action* $\alpha(e) \in G$ which acts on the current logic when the edge is selected. To focus on learning hard steps, we assume that the logical action of edges not in E_s are trivial, $\alpha|_{E_s^c} := e_G$. Let $\psi : S \rightarrow \mathcal{R}$ be an arbitrary embedding map. For a valid path $X_{0:T} \subseteq S$, we define the corresponding logic sequence $r_{0:T} \subseteq \mathcal{R}$ as

$$r_0 = \psi(X_0), \quad r_t = \alpha(X_{t-1}, X_t) \cdot r_{t-1}.$$

For example, if each state is a Boolean expression being manipulated according to certain rules, $\mathcal{R} = G = \mathbb{Z}_2$ could be used to encode the evaluation of the current expression by switching between 1 (True) and 0 (False) depending on the effect of each manipulation. G could also be taken to be a space of functions with the evaluation action $g \cdot r = g(r)$, so that the logic computes a repeated composition of functions. When the chain terminates, the final logic $r_T =: r(X_{0:T})$ is returned. Note that logical values are not unique to states and r_T depends on the entire path $X_{0:T}$.

Logic Task. Given $(X_{\text{in}}, X_{\text{out}}) \sim \mathcal{D}$, the goal is to output both a valid path $X_{0:T}$ from X_{in} to X_{out} and its logical value $r(X_{0:T})$. Since any path can be made simple by deleting loops, here we require valid paths to be simple.

To establish a rigorous distinction between the use of a search algorithm and lack thereof, we consider models consisting of a pretrained base model or *linguistic* component $\mathcal{M}_p$, responsible for learning p and generating a valid CoT, and a *reasoning* component f_θ , which predicts the answer $r(X_{0:T})$ based on (limited) information from $\mathcal{M}_p$. As in Section 3, we suppose $\mathcal{M}_p$ has perfectly learned the kernel p and can output arbitrary valid paths $\mathcal{M}_p(X_{\text{in}}, X_{\text{out}})$, solving the first part of the task. Here we do not consider the time complexity of running $\mathcal{M}_p$, which (as we have seen in Theorem 3.2) can be quite long without a search-and-improvement protocol. Thus the main task of the reasoner is to execute logical computations along a generated CoT.

In this section, we assume a stronger *uniform* lower bound in Assumption 3 on the minimum number of hard steps; otherwise, querying a single sparse edge $(X_{\text{in}}, X_{\text{out}}) \in E_s(p)$ could immediately reveal its (nontrivial) action.

Assumption 3’. For any $(X_{\text{in}}, X_{\text{out}}) \sim \mathcal{D}$ and any valid path $X_{0:T}$ with $X_0 = X_{\text{in}}$ and $X_T = X_{\text{out}}$, it holds that $\min |X_{0:T} \cap E_s| = \Omega(K)$.

We remark that this condition can be weakened to $\Omega(\log K)$ if $M \geq \Omega(K)$, in which case our results hold with $e^{-\Omega(K)}$ replaced by $K^{-\omega(1)}$.

Concept class. Define $\mathcal{P}$ the set of transition kernels on S satisfying Assumptions 1, 2 and denote the sparse edge set of $p \in \mathcal{P}$ as $E_s(p)$. The logical action α can be seen as generated by sampling $\mathcal{A} : S \times S \rightarrow G$ i.i.d. uniformly from G , then masking out all edges

not in $E^s(p)$ by setting them to e_G . Thus $\mathcal{A}$ can be regarded as a variable separate from the target p , and the logic is computed recursively as

$$r_{\mathcal{A},p}(X_0) = \psi(X_0), \quad r_{\mathcal{A},p}(X_{0:t}) = \begin{cases} \mathcal{A}(X_{t-1}, X_t) \cdot r_{\mathcal{A},p}(X_{0:(t-1)}) & (X_{t-1}, X_t) \in E_s(p) \\ r_{\mathcal{A},p}(X_{0:(t-1)}) & (X_{t-1}, X_t) \notin E_s(p). \end{cases}$$

Finally, the logic $r_{\mathcal{A},p}(X_{0:T})$ is mapped to a scalar output via a classifier $\phi : \mathcal{R} \rightarrow \{+1, -1\}$. We assume that $\mathbb{E}_{g \in G}[\phi(g \cdot r)] = 0$ for all $r \in \mathcal{R}$. The concept class is thus

$$\mathcal{H} = \{h_p \in S \times S \times G^{|S| \times |S|} : p \in \mathcal{P}, \\ h_p(X_{\text{in}}, X_{\text{out}}, \mathcal{A}) = \phi \circ r_{\mathcal{A},p}(\mathcal{M}_p(X_{\text{in}}, X_{\text{out}}))\},$$

equipped with inner product $\langle h_p, h_{p'} \rangle_{\mathcal{H}} := \mathbb{E}_{(X_{\text{in}}, X_{\text{out}}) \sim \mathcal{D}, \mathcal{A}}[h_p(X_{\text{in}}, X_{\text{out}}, \mathcal{A}) h_{p'}(X_{\text{in}}, X_{\text{out}}, \mathcal{A})]$.

5.2 A Measure of Hardness with Restricted Access

In previous sections, we have seen that pretraining $\mathcal{M}_p = \hat{p}_{\mathbf{W}}$ and running a search or distillation algorithm f_{θ} will correctly infer the underlying sparse structure. In this case, computing $r_{\mathcal{A},p}(\mathcal{M}_p(X_{\text{in}}, X_{\text{out}}))$ is trivial by concatenating actions along the identified sparse edges. In contrast, we now restrict the reasoning component's access to p by only allowing certain queries to $\mathcal{M}_p$. This makes it difficult to infer the sparse structure and true logical actions.

To understand learning with this additional (restricted) information, we propose the following generalization of the statistical query dimension (Kearns, 1998; Feldman, 2017).

Definition 5.1 (SDA: SQDIM with access). Let $\mathcal{P}$ be the set of ground truths and $\mathcal{H} = \{h_p : \mathcal{X} \rightarrow \{\pm 1\} \mid p \in \mathcal{P}\}$ the associated concept class with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$. Let $\mathcal{I}_p$ be any value or any function on $\mathcal{X}$ depending on p . Then the *statistical query dimension of $\mathcal{P}$ with access to $\mathcal{I}$* and tolerance τ is defined as

$$\text{SDA}_{\tau}(\mathcal{P}; \mathcal{I}) := \sup\{|\mathcal{P}'| : \mathcal{P}' \subseteq \mathcal{P}, \\ |\langle h_{p_1}, h_{p_2} \rangle_{\mathcal{H}}| \leq \tau, \mathcal{I}_{p_1} = \mathcal{I}_{p_2} \forall p_1 \neq p_2 \in \mathcal{P}'\}.$$

In this section, we consider $\tau = 0$ and omit its notation. Extending classical analyses (e.g., Shalev-Shwartz et al., 2017; Shamir, 2018), we prove a general limitation for gradient-based learning when additional information $\mathcal{I}_p$ is provided.

Theorem 5.2 (SQ learning with additional information). *Let f_{θ} be any parametric model of the form*

$$x \mapsto f_{\theta}(x, \mathcal{I}_p(x)).$$

Let the loss function be $L(\theta; p) := \|h_p - f_\theta\|_{\mathcal{H}}^2$ and set $\delta := (4\|\nabla f_\theta\|_{\mathcal{H}}^2 / \text{SDA}(\mathcal{P}, \mathcal{I}))^{1/3}$. Then choosing p randomly from a subset of $\mathcal{P}$, any iterative algorithm $A(\theta)$ that makes at most n queries to the δ -corrupted gradient oracle ∇L has expected loss

$$\mathbb{E}_p[L(A(\theta); p)] \geq 1 - \text{SDA}(\mathcal{P}, \mathcal{I})^{-1}$$

with probability at least $1 - n\delta$.

We only consider the squared loss in our formulation for simplicity. While squared loss only answers correlational queries, CSQ-learnability is equivalent to SQ-learnability for Boolean concepts (Bshouty and Feldman, 2001).

5.3 Results on Hardness of Logical Task

We consider four types of access to the pretrained model. Note that a *local neighborhood* of a subset $S' \subset S$ in the weighted directed graph defined by p is defined as the subgraph consisting of states reachable with a bounded number of steps from any state in S' .

- (1) **No pretraining**, $\mathcal{I}_p \equiv \emptyset$: the learner $f_\theta(X_{\text{in}}, X_{\text{out}}, \mathcal{A})$ has not been pretrained and does not receive any information on p .
- (2) **Path-only (no search)**, $\mathcal{I} \equiv \mathcal{M}$: the learner is allowed to depend on inputs $X_{\text{in}}, X_{\text{out}}$, and $\mathcal{A}$, and also the generated path $\mathcal{M}_p(X_{\text{in}}, X_{\text{out}})$. That is, the linguistic component (base model) will return a valid CoT for the input at hand, but we cannot simulate different chains from p to execute some search policy or inference algorithm.
- (3) **Local search**, $\mathcal{I} \equiv \text{nb}(\mathcal{M})$: the learner is allowed full access to a local neighborhood of $\mathcal{M}_p(X_{\text{in}}, X_{\text{out}})$ in the graph of p , including connectivity information and transition probabilities. For instance, it can flag low-probability edges as more likely to be sparse, or run bounded-length CoT from X_{in} or X_{out} .
- (4) **Full search**, $\mathcal{I} \equiv \mathcal{P}$: the learner is given full access to the entire graph of p at all times. In this case, Algorithm 2 or 4 can be used to infer $E_s(p)$ and generate CoT efficiently, and also perform the desired computation h_p .

Our main negative result states that (1)-(3) *cannot* solve the logic task with polynomial compute, and thus global search is necessary:

Theorem 5.3. $\text{SDA}(\mathcal{P}; \mathcal{P}) = 1$ and

$$\text{SDA}(\mathcal{P}; \emptyset) \geq \text{SDA}(\mathcal{P}; \mathcal{M}) \geq \text{SDA}(\mathcal{P}; \text{nb}(\mathcal{M})) \geq e^{\Omega(K)}.$$

Remark 5.4. The necessity of global information for certain learning problems (*globality barrier*) has been conjectured in Abbe et al. (2024), where the hardness of a ‘cycle task’ is proved. These results are also closely related to classical SQ-hard problems such as subset parity. The precise relationship between SDA, globality and learning is still open.